Arunabha Majumder

Mechatronics Engineer

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EDUCATION

PhD, Mechatronics and Robotics
Aalborg University, Denmark
2023 - present, thesis expected to be finished in March

MTech, Mechatronics
AcSIR - CSIR-CMERI, India
2020 - 2022

BTech, Mechanical Engineering
Siddaganga Institute of Technology, India
2015 - 2019

Senior Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad
2008, First Division with Distinction

STUDY ABROAD

National University of Singapore
Study abroad stay during my PhD
Sep - Nov 2025

SKILLS

CAD: SolidWorks (CSWA), Fusion 360 (basic, last used 2019), technical drawings

Prototyping: 3D printing, hands-on assembly

Hardware: motors, encoders, force/torque sensors, Arduino

Control and data: MATLAB/Simulink, LabVIEW, Python (basic), Qualisys motion capture and EMG

Art and design: hand sketching, painting, Adobe Illustrator, Photoshop, Procreate

ACHIEVEMENTS

Best Paper Award, IFToMM ISRM 2026

Best Major Project Award, 2019

TATA Technologies Prayatna grant

GATE qualified

LANGUAGES

English (C1), Bengali (native), Hindi, German (willing to learn)

REFERENCES

Prof. Shaoping Bai
Aalborg University, sbh@mp.aau.dk

Assoc. Prof. Lars Rosgaard Jensen
Head of Section and Head of Labs, AAU

Assoc. Prof. Anderson de Souza Castelo Oliveira
Materials and Production, AAU

PROFILE

I am a mechatronics engineer who likes building real robot hardware and getting all of it to work together: the mechanics, the motors and sensors, the control loop and the data side. Over the last 6 years I have designed exoskeletons and robots, put them together, integrated the electronics, and run experiments with real people. I like small teams where I can work across the whole hardware stack, learn from what goes wrong in the lab, and keep making the system simpler for the person who uses it.

EXPERIENCE

Research Assistant

Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton: a variable stiffness mechanism (spring pretensioned by a worm gear) working in parallel with a 6 Nm motor. It uses over 25% less motor energy (Best Paper Award, IFToMM ISRM 2026).
- Optimised the design with a genetic algorithm, modelled it in SolidWorks, 3D printed the first parts and had the final parts manufactured.
- Integrated the motor, encoder and force/torque sensors and measured its stiffness on a physical setup.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance.
- Arm impedance study with the 5-bar robot (6 participants): Qualisys cameras tracked the arm, and I implemented an Arduino-based trigger to sync the robot's motion with the recording.
- Exoskeleton study: collected EMG data from 12 participants with the Qualisys data acquisition system.
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit

CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit. Designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

Bachelor's Projects

Siddaganga Institute of Technology, India | *2017 - 2019*

- Built an arm exoskeleton and exo glove for a paralysed arm: CAD, machining, assembly and testing.
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

PUBLICATIONS

- Hybrid actuated shoulder exoskeleton, IFToMM ISRM 2026 (Springer), Best Paper Award.
- Neural network-based control of a pneumatic artificial muscle, IEEE CONECCT 2022.

PORTFOLIO

Videos and results from my work: arunabha22.github.io/arunabha-majumder